Assembly Manual

Licence plate detection with Jetson Nano

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Submission date:	October 14, 2025

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Acronyms

1 General Assembly Hints

Before beginning the assembly of the License Plate Detection System, please read the following important guidelines. These instructions ensure a safe, efficient, and error-free setup process for the Jetson Nano hardware and its connected components.

Preparation and Safety Measures

- **Work Environment:** Assemble all components on a clean, dry, and non-conductive surface. Ensure proper lighting and ventilation to avoid overheating during operation.
- **Electrostatic Discharge (ESD):** Handle the Jetson Nano and camera only by their edges. Use an antistatic wrist strap or touch a grounded metal object before handling any electronics.
- **Cable Management:** Arrange all cables neatly and ensure that they are not under tension. Avoid crossing power and data cables where possible.
- **Component Orientation:** Check the correct direction of all connectors (micro-SD, USB, HDMI, and power) before insertion. Do not apply excessive force.
- **Power Supply:** Use only the official 5 V - 4 A adapter. Always disconnect power before adding or removing components.

Table 1.1: Assembly Preparation Checklist

Item	**Description / Action Required**
Work Surface	Clean, dry, non-conductive, and well-lit area
Power Adapter	Verified 5V - 4A supply (official Jetson Nano adapter)
SD Card	32 GB card pre-flashed with JetPack 5.1.2 (Ubuntu 20.04)
Display and Peripherals	HDMI monitor, USB keyboard, and mouse ready
Camera Module	Microsoft LifeCam Show (USB) connected but protected with lens cover
Network Connection	Ethernet or Wi-Fi adapter available for first boot
Static Safety	ESD wrist strap or grounding point nearby

Ventilation and Safety

- Place the Jetson Nano on a hard, flat surface with sufficient airflow around the heat sink.
- Do not operate the system inside a closed enclosure without ventilation holes.
- Avoid touching the heat sink during or immediately after operation, as it may become hot.
- If long-duration tests are performed, use an external cooling fan for optimal performance.

Post-Assembly Verification

After completing all connections, verify the following points:

1. The power LED on the Jetson Nano turns on when the adapter is connected.
2. The HDMI display output is visible on the monitor.
3. The USB camera is detected and functioning properly.
4. All cables and connectors are securely attached.
5. No abnormal heat or noise occurs during operation.

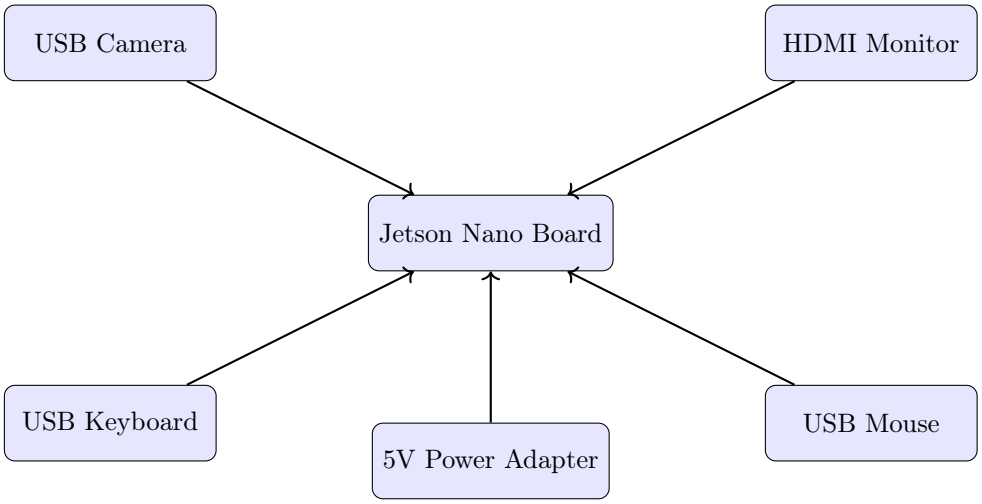

Figure: Assembly layout for Jetson Nano hardware setup.

Following these general assembly hints ensures reliable installation, minimizes hardware risks, and provides a stable foundation for further software configuration and testing.

2 Main Function

The assembled unit implements real-time license plate acquisition on the NVIDIA Jetson Nano. The USB camera captures frames, the Jetson performs detection and optical character recognition, and results can be visualized on the HDMI display. This chapter describes the overall function, physical interconnects, and power path.

Overview

- **Compute Core:** Jetson Nano (4 GB) processes video frames locally.
- **Image Acquisition:** Microsoft LifeCam Show (USB) delivers the input stream.
- **User I/O:** HDMI monitor for display; USB keyboard and mouse for setup.
- **Storage:** 32 GB microSD with JetPack 5.1.2.
- **Power:** 5 V 4 A DC adapter.

Table 2.1: Interfaces Used by the Assembly

Subsystem	**Jetson Port**	**Notes**
USB Camera	USB 2.0/3.0 Type-A	Connect directly to a USB port on Jetson Nano.
Display	HDMI	Use a short high-quality HDMI cable.
User Input	USB Type-A	Keyboard and mouse on separate ports or via USB hub.
Network (optional)	Ethernet (RJ45)	Recommended for first setup and package updates.
Power	DC Barrel Jack	Official 5 V 4 A adapter (5 V DC).
Storage	microSD slot	32 GB microSD with JetPack 5.1.2.

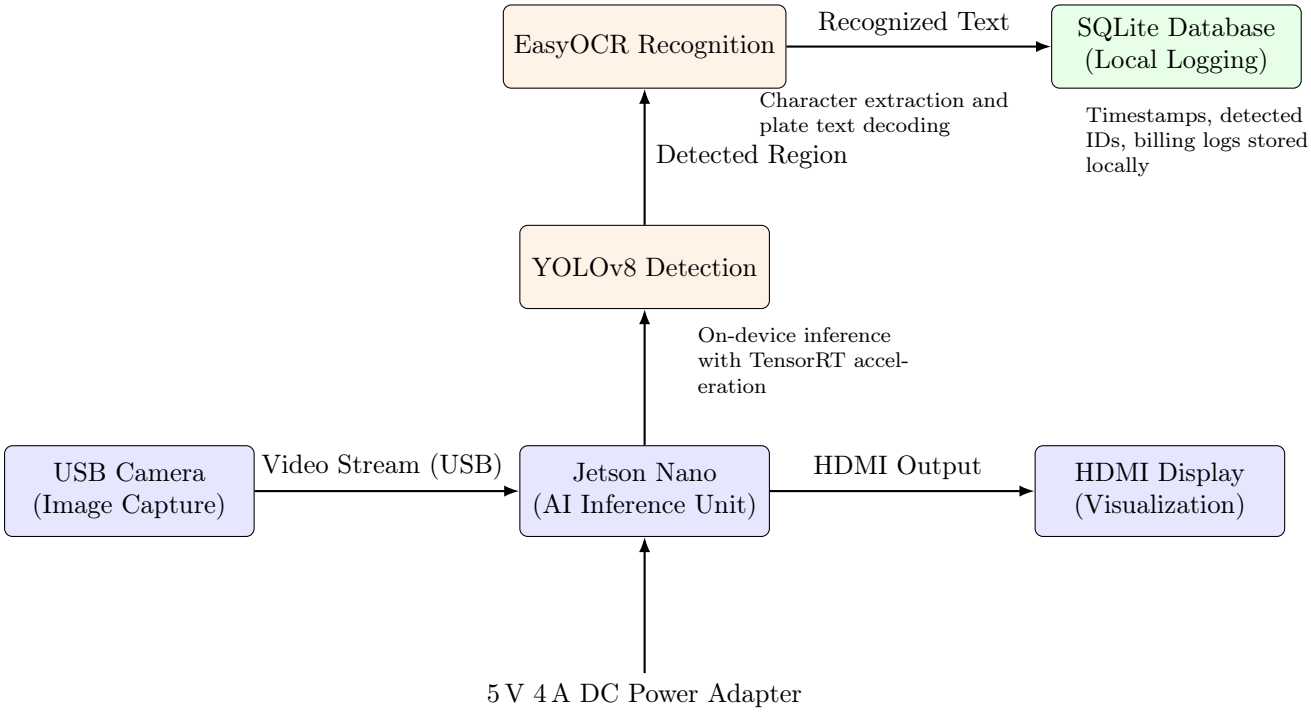

Figure 2.1: Overall license plate detection workflow implemented on Jetson Nano: from image capture to OCR and local database logging.

This diagram visually represents the complete end-to-end workflow of the system and the logical interaction between its key modules. It begins with the **USB camera**, which continuously captures live video frames from the environment. These frames are transmitted through the USB interface to the **Jetson Nano**, which serves as the system's central AI processing unit. Inside the Nano, the incoming image stream is processed by a pre-trained **YOLOv8 model** optimized for edge inference using TensorRT acceleration. Once a vehicle license plate is detected, the corresponding image region is forwarded to the **EasyOCR recognition layer**, which extracts alphanumeric text with high precision even under varying lighting conditions. The recognized plate numbers, along with timestamps, detection confidence scores, and event details, are then stored in the local **SQLite database** for persistent on-device logging and later retrieval. The output from the Jetson Nano is displayed on an **HDMI monitor** for live visualization of detection results, providing both verification and debugging feedback for the operator. The entire pipeline is powered by a regulated 5 V, 4 A DC adapter that ensures stable GPU performance, thermal efficiency, and uninterrupted operation during continuous video inference. This workflow clearly outlines the system's real-time nature—combining data acquisition, intelligent detection, optical recognition, and data management—all within a compact and power-efficient embedded platform.

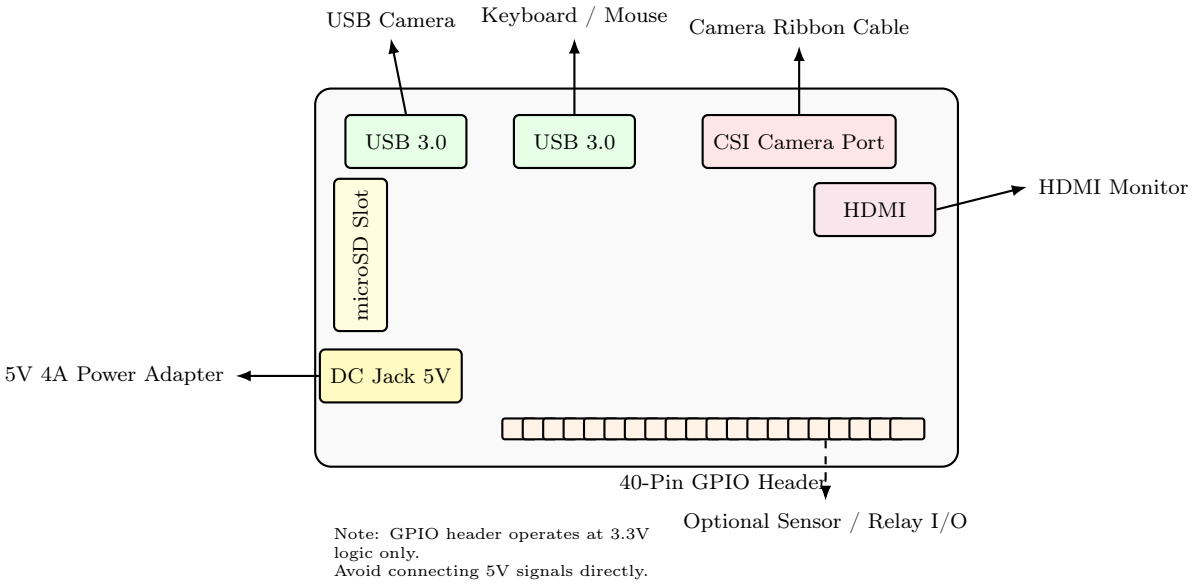

Figure 2.2: Top-view Jetson Nano hardware layout showing only the active connections used in the license plate detection system — USB camera, HDMI display, keyboard/mouse, CSI camera, GPIO, microSD, and 5V DC power.

The above top-view layout illustrates the exact hardware configuration of the assembled system. It shows how the USB camera, HDMI monitor, keyboard, and mouse are connected to the Jetson Nano through their respective ports, while power is supplied via a 5 V DC adapter. The CSI camera interface and GPIO header are highlighted for potential future extensions, and the microSD card slot provides storage for the operating system and project files. Arrows indicate the direction of power and data flow, ensuring that the physical connections match the functional workflow shown in the previous diagram.

Table 2.2: Cable and Connector Checklist

Connection	**Checkpoints**
USB Camera → Jetson	Firm seating, no strain; short cable preferred.
HDMI → Monitor	Cable fully inserted; monitor set to correct input source.
Keyboard / Mouse → Jetson	Separate ports or powered hub; verify LEDs (if present).
Ethernet → Router/Switch	Link LEDs active; stable connection for first boot.
DC Adapter → Jetson	Official 5 V 4 A; avoid loose barrel jack.

3 List Of Parts

Selecting and documenting the correct hardware components is crucial to ensuring reliable and repeatable results in embedded AI systems. This section provides the full bill of materials (BOM) used in the assembly of the license plate detection prototype. The listed components were chosen based on criteria such as power efficiency, availability, community support, and proven use in prior edge-AI research [NVI24; SEG22; War19; JCQ23].

Table 3.1: Detailed Component List and Functions

Part Name	**Technical Specification**	**Function in System**	**Qty**
Jetson Nano (4 GB, B01)	Quad-core ARM Cortex-A57, 128-core Maxwell GPU; 40-pin GPIO	Runs YOLOv8 detection + EasyOCR; central compute	1
microSD Card	32 GB UHS-I, Class 10; JetPack 5.1.2 image	OS boot / model and data storage	1
USB Camera	UVC compliant; 1080p@30 FPS (e.g., LifeCam Show)	Captures live plate images	1
HDMI Display	1080p monitor with HDMI input	Live visualization and debugging	1
USB Keyboard + Mouse	Wired or 2.4 GHz dongle	Local setup and control	1 set
DC Power Supply	5 V, 4 A regulated, barrel jack	Stable power for sustained inference	1
HDMI Cable	High-speed, ≈1 m	Connects Nano to display	1
USB Cables	USB-A to device (camera/peripherals)	Camera/peripheral connectivity	1–2
Enclosure (optional)	Jetson-compatible case or reComputer shell	Physical protection and airflow	1
GPIO Jumper Set (optional)	Male–Female, ≈20 cm	Future sensors / I/O prototyping	as req.

The above table summarizes the core and supporting components required for the embedded vision pipeline. Each item was validated under load to ensure sustained inference performance at approximately 10–15 FPS. Using the official 5 V/4 A NVIDIA adapter prevents under-voltage throttling of the GPU, as highlighted in [GBMP23].

Cable lengths are deliberately kept below 1.2 m to reduce data latency and power loss.

Table 3.2: Optional Accessories and Spares

Accessory	**When Useful**	**Notes**
Powered USB Hub	When multiple peripherals exceed onboard current limit	Reduces system stress on 5V rail
USB Fan / Heatsink	Continuous inference runs >30 min	Improves thermal margin up to 15% [LRF22]
Tripod / Camera Mount	Required for fixed surveillance perspective	Maintains plate alignment consistency
Spare microSD	Quick OS re-image / recovery point	Ensures reproducibility of environment
Cable Ties / Labels	Recommended for lab safety	Prevents mechanical strain and confusion

Accessories in Table 2 are recommended for robustness and testing consistency. The fan or heatsink improves long-term thermal stability, which directly impacts inference accuracy on edge devices [LRF22]. Maintaining fixed camera geometry via a tripod minimizes variation in dataset capture.

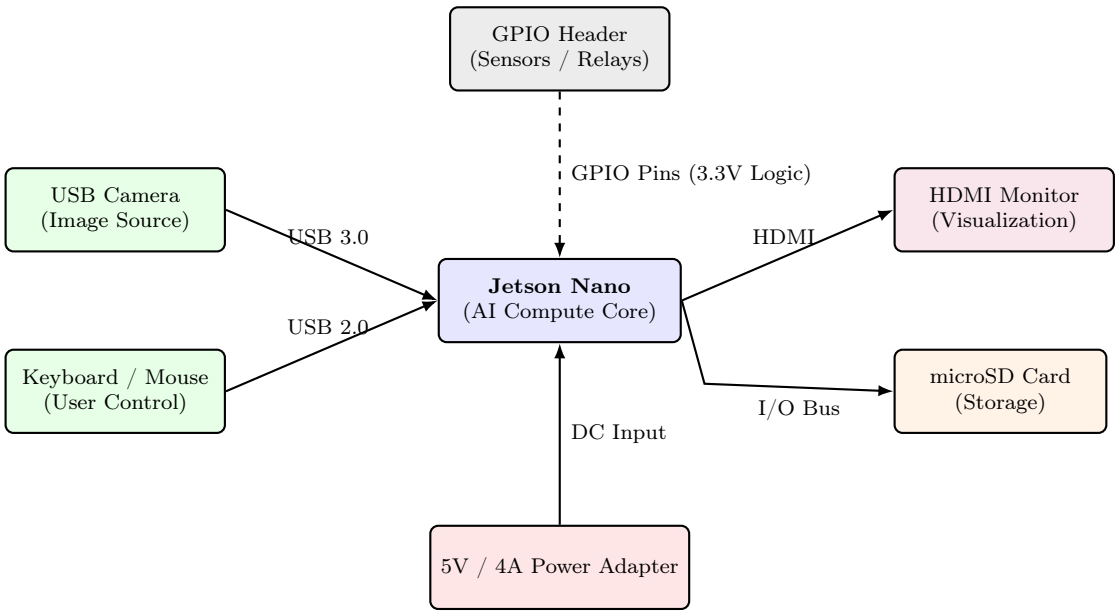

Figure 3.1: System assembly layout showing core Jetson Nano connections with color-coded layers for input, output, power, and optional expansion.

This diagram demonstrates the modular architecture of the embedded system. Each component is color-coded by function, enabling easy identification during assembly and inspection. Such layered representations are commonly used in embedded AI system documentation [HPY22; SEG22]. The dashed GPIO link emphasizes scalability, supporting future hardware upgrades such as relay control, vehicle sensors, or IoT integration.

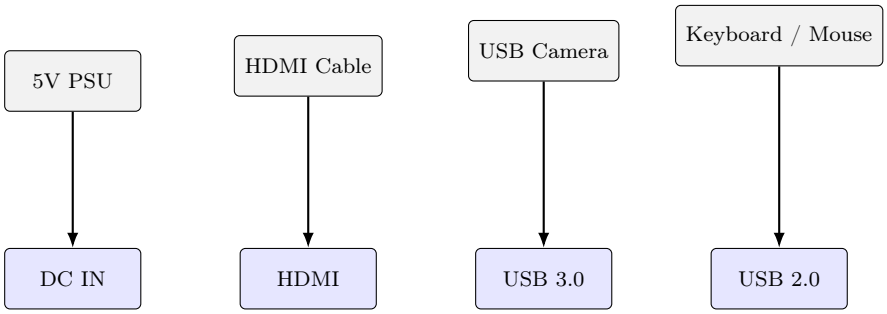

Figure 3.2: Cable pack to port mapping used during assembly. Each arrow indicates the correct connection for power, display, camera, and peripherals.

The cable mapping provides a simplified "port-to-cable" visualization, minimizing wiring errors during setup. This approach reflects the design conventions used in modern hardware documentation to improve assembly accuracy and reproducibility [Ric21; HLZ22]. Consistent use of short, labeled cables improves electromagnetic compatibility (EMC) and reduces signal degradation, as noted by [KBS21; ZWC23].

4 Setup

This chapter walks you through preparing the microSD card, assembling the hardware, completing the first boot configuration, and validating the camera and display. Follow the steps in order and use the checklists to prevent common mistakes.

4.1 Before You Begin

Workspace and Safety

- Work on a clean, non-conductive surface with good lighting.
- Avoid electrostatic discharge (ESD): touch a grounded metal object before handling the board.
- Keep cables short and routed to avoid strain on connectors.

You Will Need

- NVIDIA Jetson Nano (4 GB, B01).
- 32 GB microSD card (UHS-I, Class 10) prepped with JetPack 5.1.2.
- Official 5 V, 4 A DC power adapter (barrel jack).
- USB camera (UVC compliant; e.g., Microsoft LifeCam Show).
- HDMI monitor + HDMI cable.
- USB keyboard and mouse (wired or 2.4 GHz dongle).

  Tip. Faster microSD media (high sustained write, UHS-I) noticeably improves system responsiveness and logging performance.

4.2 Prepare the microSD Card

Download and Flash

1. Download the Jetson Nano image for JetPack 5.1.2 from NVIDIA.
2. Use a flashing tool (e.g., Raspberry Pi Imager or balenaEtcher) to write the image to the 32 GB microSD card.

3. After flashing completes, safely eject the card.

General/Images/sd_flashing_overview.png

Figure 4.1: Flashing JetPack image to microSD using a cross-platform imager.

> **Note.** If you maintain a "known-good" cloned card, you can swap cards to recover quickly without re-imaging.

4.3 Hardware Assembly

Install and Connect

1. Insert the flashed microSD card into the Jetson Nano microSD slot.
2. Connect the HDMI cable from the Jetson to the monitor.
3. Plug in the USB camera to a USB 3.0 port (prefer the blue receptacle).
4. Connect keyboard and mouse (separate USB ports, or a receiver dongle).

5. *Do not* power the board yet.

General/Images/jetson_top_layout.jpg

Figure 4.2: Jetson Nano top layout with key ports. Use short, good-quality cables.

General/Images/wiring_overview.jpg

Figure 4.3: Wiring overview: USB camera, HDMI to monitor, keyboard/mouse, and DC barrel power.

Warning. Only use a regulated 5 V, 4 A adapter. Underrated supplies can cause throttling or brownouts.

4.4 First Boot Configuration

Power On and Wizard

1. Connect the 5 V adapter to the DC barrel jack to power on.
2. Complete the first-boot wizard: locale, keyboard, user account, and display settings.
3. Verify the monitor is set to the correct HDMI input.

Recommended System Settings

From a terminal:

```
# Performance mode and clocks (Jetson Nano)
sudo nvpmodel -q         # Check current mode
sudo nvpmodel -m 0       # Max performance profile
sudo jetson_clocks       # Lock clocks for consistent benchmarking
```

(Optional) Create swap to improve stability during large installs:

```
# Create a 2 GB swapfile
sudo fallocate -l 2G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
# Make persistent:
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
```

4.5 Camera and Display Validation

Detect the Camera

```
lsusb
v4l2-ctl --list-devices
v4l2-ctl --list-formats-ext -d /dev/video0
```

Live Preview (USB UVC Camera)

```
# GStreamer preview (generic UVC):
gst-launch-1.0 v4l2src device=/dev/video0 ! videoconvert ! autovideosink
```

HDMI Output Sanity Check

```
xrandr            # Confirm HDMI mode
```

If the monitor remains blank, reseat the HDMI cable and ensure the correct input source is selected.

4.6 Software Prerequisites (Minimal)

Base Tools

```
sudo apt update
sudo apt install -y python3-pip python3-venv git \
gstreamer1.0-tools v4l-utils
```

Python Environment (Optional but Recommended)

```
python3 -m venv ~/lpd-venv
source ~/lpd-venv/bin/activate
pip install --upgrade pip
# install your project requirements afterwards, e.g.:
# pip install opencv-python numpy onnxruntime easyocr
```

4.7 Quick Setup Checklist

Table 4.1: Setup Checklist

Task	**Status / Notes**
microSD flashed with JetPack 5.1.2	_____
HDMI connected, monitor input selected	_____
USB camera detected as `/dev/video0`	_____
Keyboard/mouse functional on login	_____
Performance mode enabled (`nvpmodel -m 0`)	_____
Live camera preview verified (GStreamer)	_____

4.8 Troubleshooting Hints (Setup Phase)

- **No display:** Check HDMI input source; try a different cable; test at

5 Functions

6 Maintenance

7 Safety Guidelines

8 Troubleshooting

9 Add Ons

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